Cairo University

Faculty of Engineering

Electronics and Electrical Communications Engineering

Assignment 1

ELC 4028 – Artificial Neural Networks

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PART 1: Regression Problems

1.1 Problem 1: Insurance Company

The numbers of insured persons y with an insurance company for the years 1987 to 1996 are given. We let x represent the number of years since 1987 ($x = 0$ for 1987, $x = 9$ for 1996).

1.1.1 Dataset

x (years since 1987)	Year	y (insured persons)
0	1987	12,400
1	1988	10,900
2	1989	10,000
3	**1990**	**1,050 (OUTLIER)**
4	1991	9,500
5	1992	8,900
6	1993	8,000
7	1994	7,800
8	1995	7,600
9	1996	7,200

Table 1: Insurance dataset (1987–1996)

** Value 1050 at $x = 3$ (year 1990) is the outlier — likely a data entry error or anomaly.*

1.1.2 Case A: WITH Outlier

Scatter Plot & Outlier Identification A scatter plot of y (number of insured persons) versus x (years since 1987) reveals a generally decreasing trend from $\approx$ 12,400 down to $\approx$ 7,200. However, one data point stands out sharply:

- $x = 3$ (year 1990): $y = 1050$

This value is dramatically lower than the surrounding data, which are all within approximately 7,200 to 12,400. The sharp drop breaks the otherwise smooth downward trend. This observation is therefore identified as a clear outlier, most likely due to a data entry error (e.g. 10,500 instead of 1,050).

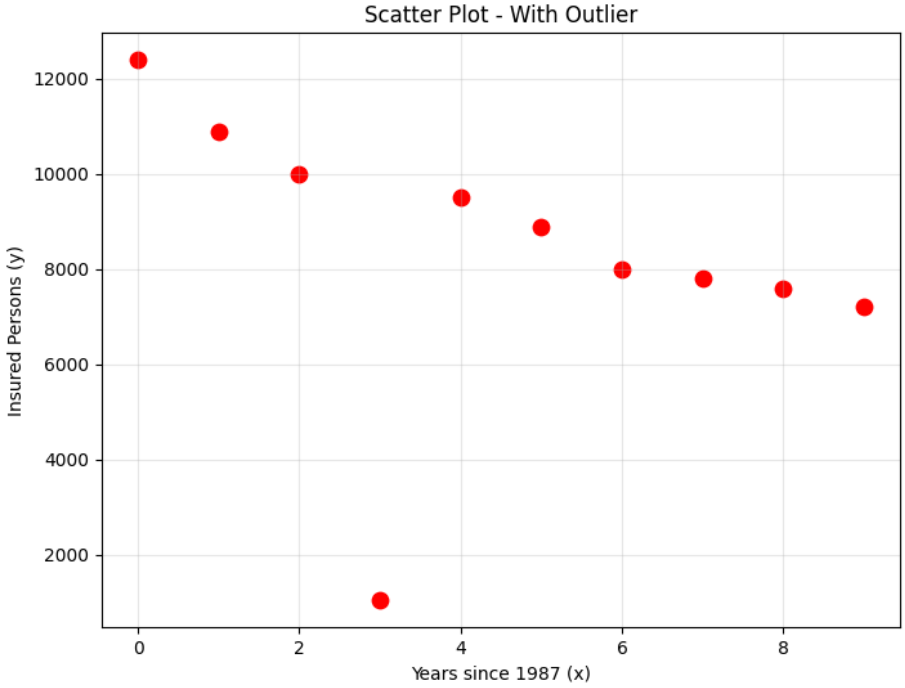


Figure 1: Scatter plot of original dataset identifying the outlier at $x = 3$ ($y = 1050$)

Regression Models (WITH Outlier) We apply both OLS and matrix-form methods to compute linear model parameters, then use the matrix form for quadratic and cubic models.

Linear Model Hand Analysis (OLS Method)

The equation for the linear model is:

$$\hat{y}_i = \beta_0 + \beta_1 x_i \tag{1}$$

The key formulas:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} x_i y_i - \frac{\left(\sum x_i\right)\left(\sum y_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum x_i\right)^2}{n}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \tag{2}$$

Table 2: Data table for hand computation (WITH outlier)

Year	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1987	0	12,400	0	153,760,000	0
1988	1	10,900	1	118,810,000	10,900
1989	2	10,000	4	100,000,000	20,000
1990	3	1,050	9	1,102,500	3,150
1991	4	9,500	16	90,250,000	38,000
1992	5	8,900	25	79,210,000	44,500
1993	6	8,000	36	64,000,000	48,000
1994	7	7,800	49	60,840,000	54,600
1995	8	7,600	64	57,760,000	60,800
1996	9	7,200	81	51,840,000	64,800
Σ	**45**	**83,350**	**285**	**777,572,500**	**344,750**

$n = 10, \quad \bar{y} = 83350/10 = 8335, \quad \bar{x} = 45/10 = 4.5$

$$\hat{\beta}_1 = \frac{344{,}750 - \dfrac{45 \times 83{,}350}{10}}{285 - \dfrac{45^2}{10}} = -367.57$$

$$\hat{\beta}_0 = 8335 + 367.57 \times 4.5 = 9989.09$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 9989.09 \\ -367.576 \end{pmatrix}$$

Linear equation:

$$\hat{y}_i = 9989.09 - 367.576\, x \tag{3}$$

R² Verification Table

$$\bar{y} = \frac{\sum y_i}{n} = \frac{83350}{10} = 8335$$

Table 3: R^2 verification for linear model (WITH outlier)

Year	x_i	y_i	$\hat{y}_i = \beta_0 + \beta_1 x_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	9,989.1	5,812,483	16,524,225
1988	1	10,900	9,621.5	1,634,524	6,579,225
1989	2	10,000	9,253.9	556,606	2,772,225
1990	3	1,050	8,886.4	61,408,600	53,071,225
1991	4	9,500	8,518.8	962,777	1,357,225
1992	5	8,900	8,151.2	560,683	319,225
1993	6	8,000	7,783.6	46,813	112,225
1994	7	7,800	7,416.1	147,410	286,225
1995	8	7,600	7,048.5	304,169	540,225
1996	9	7,200	6,680.9	269,455	1,288,225
Σ	45	83,350	83,350	**71,703,515**	**82,850,250**

$$R^2_{\text{linear}} = \frac{\text{SS}_{yy} - \text{SSE}}{\text{SS}_{yy}} = \frac{82{,}850{,}250 - 71{,}703{,}515}{82{,}850{,}250} = 0.1345$$

Output Graph

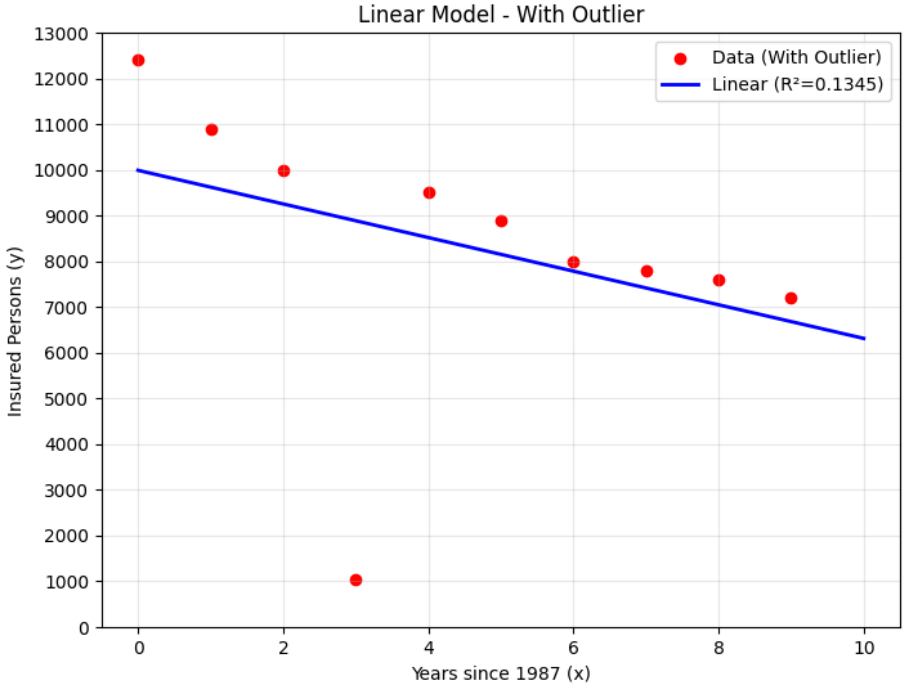


Figure 2: Linear Model of Original Data with Outlier

Simulation Results

```
Linear Model:
  Equation: y = 9989.09 + (-367.58)*x^1
  R^2: 0.1345
```

Figure 3: Simulation Results of Linear Model of Original Data with Outlier

Quadratic Model The equation for the quadratic model:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 \tag{4}$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 & 0^2 \\ 1 & 1 & 1^2 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 4 & 4^2 \\ 1 & 5 & 5^2 \\ 1 & 6 & 6^2 \\ 1 & 7 & 7^2 \\ 1 & 8 & 8^2 \\ 1 & 9 & 9^2 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 11627.73 \\ -1596.55 \\ 136.55 \end{pmatrix}$$

Quadratic equation:

$$\hat{y}_i = 11627.73 - 1596.55\,x + 136.55\,x^2 \tag{5}$$

R² Verification Table

Table 4: R^2 verification for quadratic model (WITH outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	11,627.7	596,405	16,524,225
1988	1	10,900	10,167.7	536,223	6,579,225
1989	2	10,000	8,980.8	1,038,701	2,772,225
1990	3	1,050	8,067.0	49,238,925	53,071,225
1991	4	9,500	7,426.4	4,299,969	1,357,225
1992	5	8,900	7,058.8	3,390,064	319,225
1993	6	8,000	6,964.3	1,072,639	112,225
1994	7	7,800	7,143.0	431,710	286,225
1995	8	7,600	7,594.7	28	540,225
1996	9	7,200	8,319.5	1,253,377	1,288,225
Σ	45	83,350	83,350	**61,858,042**	**82,850,250**

$$R^2_{\text{quadratic}} = \frac{82{,}850{,}250 - 61{,}858{,}042}{82{,}850{,}250} = 0.2534$$

Output Graph

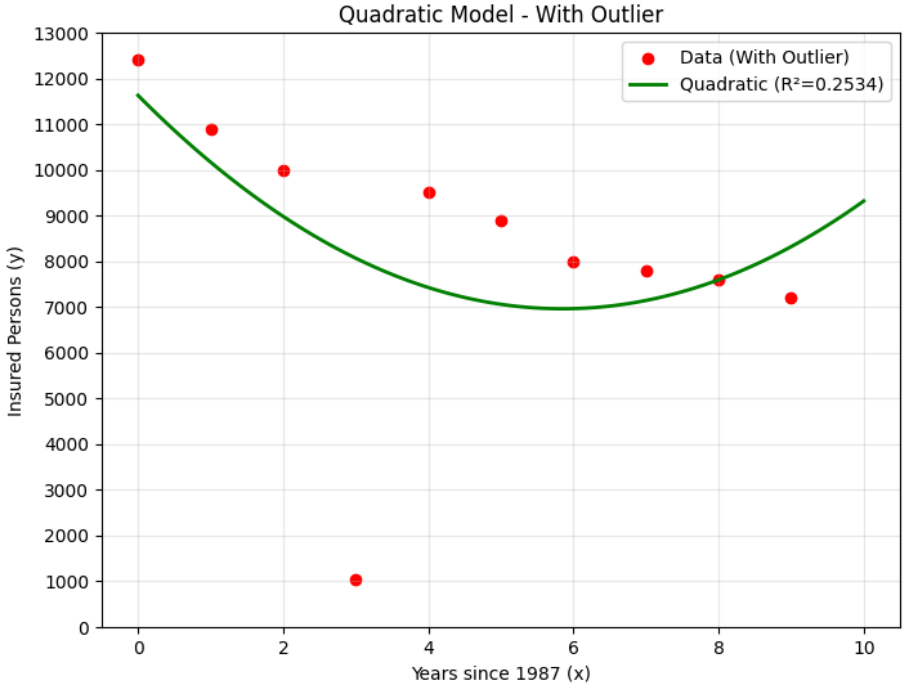


Figure 4: Quadratic Model of Original Data with Outlier

Simulation Results

```
Quadratic Model:
  Equation: y = 11627.73 + (-1596.55)*x^1 + (136.55)*x^2
  R^2: 0.2534
```

Figure 5: Simulation Results of Quadratic Model of Original Data with Outlier

Cubic Model The equation for the cubic model:

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 \tag{6}$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 & 0^2 & 0^3 \\ 1 & 1 & 1^2 & 1^3 \\ 1 & 2 & 2^2 & 2^3 \\ 1 & 3 & 3^2 & 3^3 \\ 1 & 4 & 4^2 & 4^3 \\ 1 & 5 & 5^2 & 5^3 \\ 1 & 6 & 6^2 & 6^3 \\ 1 & 7 & 7^2 & 7^3 \\ 1 & 8 & 8^2 & 8^3 \\ 1 & 9 & 9^2 & 9^3 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 1050 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 13113.15 \\ -4313.93 \\ 932.31 \\ -58.95 \end{pmatrix}$$

Cubic equation:

$$\hat{y}_i = 13113.15 - 4313.93\, x_i + 932.31\, x_i^2 - 58.95\, x_i^3 \tag{7}$$

R² Verification Table

Table 5: R^2 verification for cubic model (WITH outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	13,113.1	508,578	16,524,225
1988	1	10,900	9,672.6	1,506,542	6,579,225
1989	2	10,000	7,743.0	5,094,123	2,772,225
1990	3	1,050	6,970.7	35,054,266	53,071,225
1991	4	9,500	7,002.0	6,240,214	1,357,225
1992	5	8,900	7,483.2	2,007,341	319,225
1993	6	8,000	8,060.7	3,684	112,225
1994	7	7,800	8,380.8	337,334	286,225
1995	8	7,600	8,089.8	239,940	540,225
1996	9	7,200	6,834.1	133,864	1,288,225
Σ	45	83,350	83,350	**51,125,885**	**82,850,250**

$$R^2_{\text{cubic}} = \frac{82{,}850{,}250 - 51{,}125{,}885}{82{,}850{,}250} = 0.3829$$

Output Graph

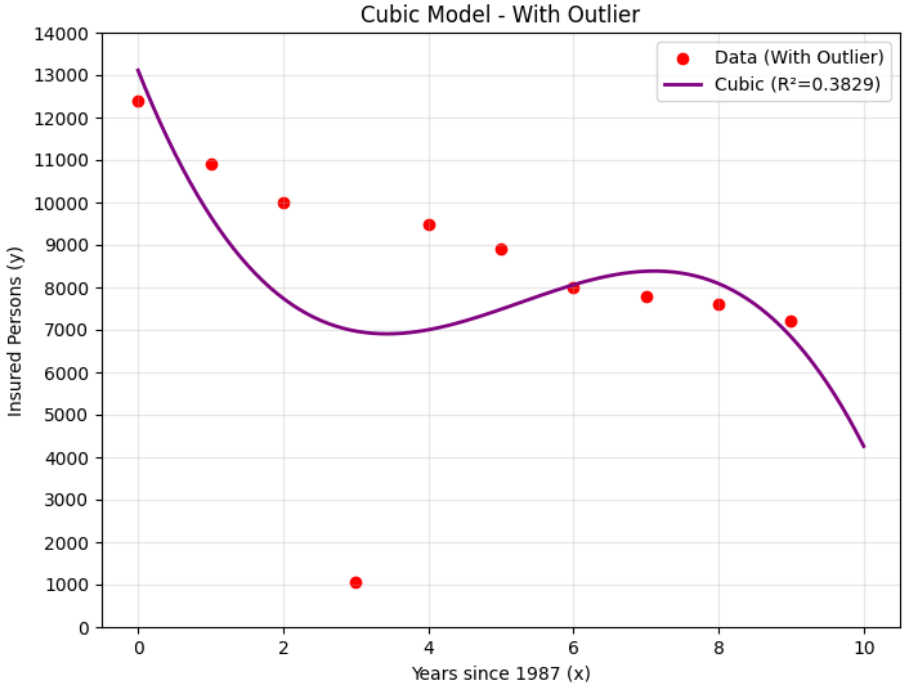


Figure 6: Cubic Model of Original Data with Outlier

Simulation Results

```
Cubic Model:
  Equation: y = 13113.15 + (-4313.93)*x^1 + (932.31)*x^2 + (-58.95)*x^3
  R^2: 0.3829
```

Figure 7: Simulation Results of Cubic Model of Original Data with Outlier

Table 6: Model comparison – WITH outlier

Model	**Equation**	R^2
Linear	$\hat{y}_i = 9989.09 - 367.576\,x_i$	0.1345
Quadratic	$\hat{y}_i = 11627.73 - 1596.55\,x_i + 136.55\,x_i^2$	0.2534
Cubic	$\hat{y}_i = 13113.15 - 4313.93\,x_i + 932.31\,x_i^2 - 58.95\,x_i^3$	0.3829

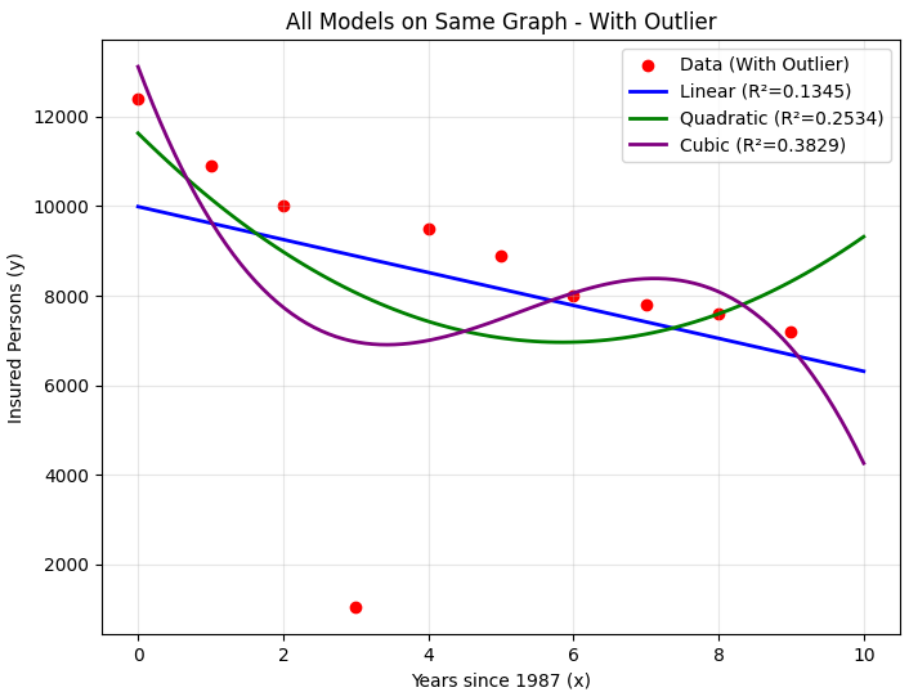


Figure 8: Comparison between three Models of Original Data with Outlier

Comparison Between Three Models (WITH Outlier) **Comment:** All R^2 values are low (< 0.50) due to the heavy influence of the outlier at $x = 3$ ($y = 1050$). The cubic model yields the highest R^2.

1.1.3 Case B: WITHOUT Outlier

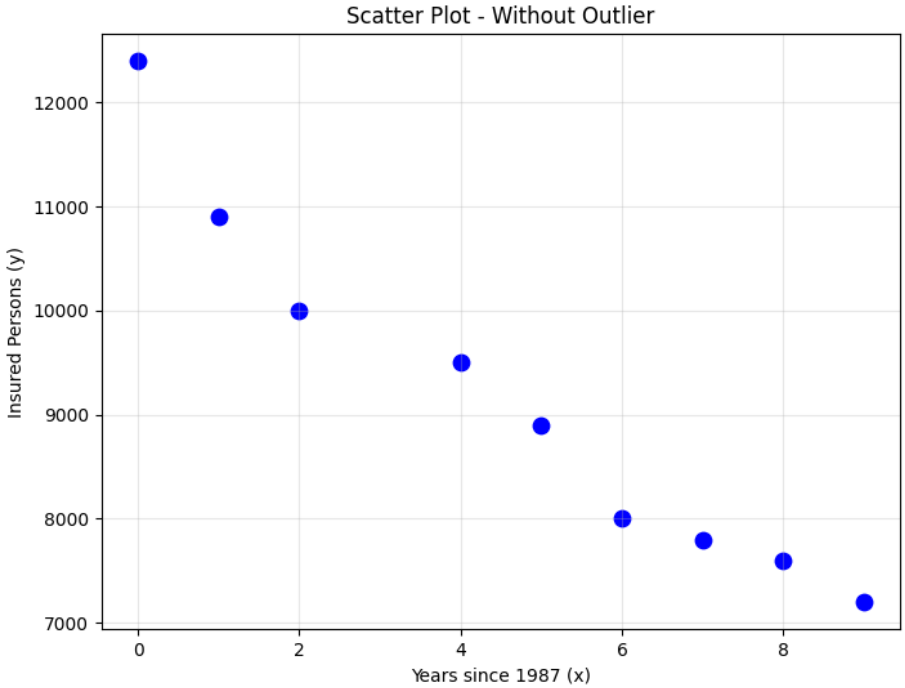


Figure 9: Scatter Plot of Cleaned Data without Outlier

Outlier Removal and Scatter Plot of Cleaned Data

Regression Models (WITHOUT Outlier) After removing the outlier at $x = 3$ ($y = 1{,}050$), the cleaned dataset contains 9 data points. We now repeat the regression analysis using the same methods.

Linear Model Hand Analysis (OLS Method)

$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

Table 7: Data table for hand computation (WITHOUT outlier)

Year	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1987	0	12,400	0	153,760,000	0
1988	1	10,900	1	118,810,000	10,900
1989	2	10,000	4	100,000,000	20,000
1991	4	9,500	16	90,250,000	38,000
1992	5	8,900	25	79,210,000	44,500
1993	6	8,000	36	64,000,000	48,000
1994	7	7,800	49	60,840,000	54,600
1995	8	7,600	64	57,760,000	60,800
1996	9	7,200	81	51,840,000	64,800
Σ	**42**	**82,300**	**276**	**776,470,000**	**341,600**

$n = 9, \quad \bar{y} = 82300/9 = 9{,}144.44, \quad \bar{x} = 42/9 = 4.667$

$$\hat{\beta}_1 = \frac{341{,}600 - \dfrac{42 \times 82{,}300}{9}}{276 - \dfrac{42^2}{9}} = -530.83$$

$$\hat{\beta}_0 = 9{,}144.44 + 530.83 \times 4.667 = 11{,}621.67$$

Hand Analysis (Matrix Form)

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}, \quad Y = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} = \begin{pmatrix} 11{,}621.67 \\ -530.83 \end{pmatrix}$$

Linear equation:

$$\hat{y}_i = 11{,}621.67 - 530.83\, x$$

R^2 Verification Table

$$\bar{y} = \frac{82300}{9} = 9{,}144.44$$

Table 8: R^2 verification for linear model (WITHOUT outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	11,621.7	605,803	10,598,642
1988	1	10,900	11,090.8	36,417	3,081,975
1989	2	10,000	10,560.0	313,600	731,975
1991	4	9,500	9,498.3	3	126,420
1992	5	8,900	8,967.5	4,556	59,753
1993	6	8,000	8,436.7	190,678	1,309,753
1994	7	7,800	7,905.8	11,201	1,807,531
1995	8	7,600	7,375.0	50,625	2,385,309
1996	9	7,200	6,844.2	126,617	3,780,864
Σ	42	82,300	82,300.0	**1,339,500**	**23,882,222**

$$R^2_{\text{linear}} = \frac{23{,}882{,}222 - 1{,}339{,}500}{23{,}882{,}222} = 0.9439$$

Output Graph

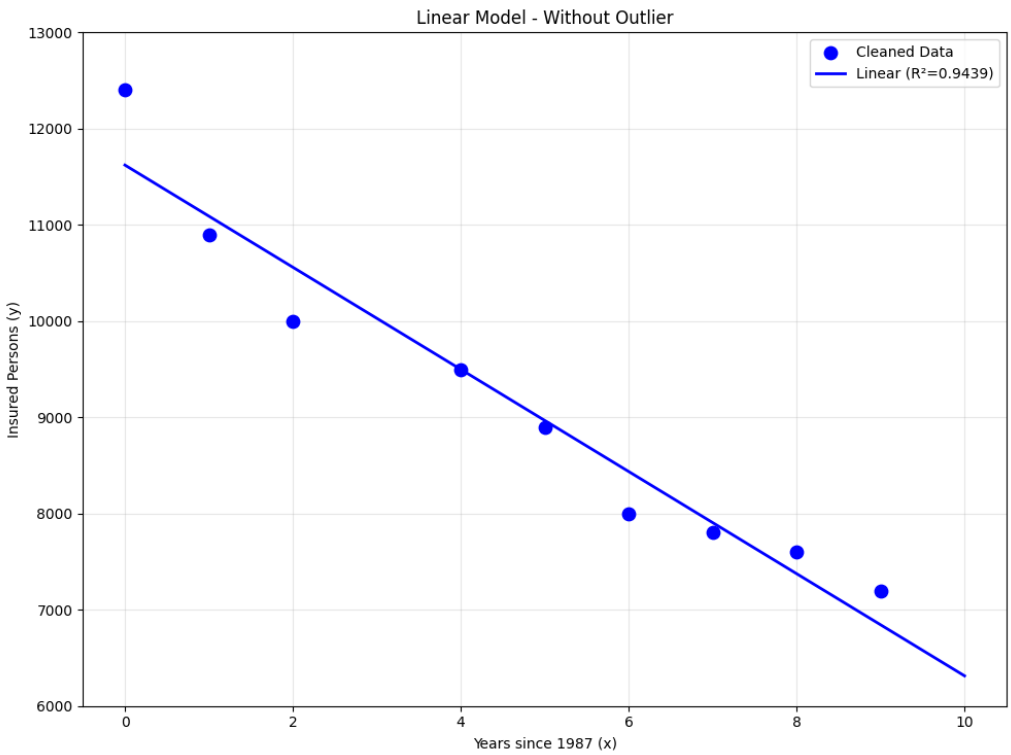


Figure 10: Linear Model of Cleaned Data without Outlier

Simulation Results

```
Linear Model:
  Equation: y = 11621.67 + (-530.83)*x^1
  R^2: 0.9439
```

Figure 11: Simulation Results of Linear Model of Cleaned Data without Outlier

Quadratic Model

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$$

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T Y, \quad \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 12{,}024.17 \\ -863.13 \\ 37.44 \end{pmatrix}$$

Quadratic equation:

$$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$$

R^2 Verification Table

$$\bar{y} = 82{,}300/9 = 9{,}144.44$$

Table 9: R^2 verification for quadratic model (WITHOUT outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	12,024.2	141,248	10,598,642
1988	1	10,900	11,198.5	89,090	3,081,975
1989	2	10,000	10,447.7	200,411	731,975
1991	4	9,500	9,170.7	108,429	126,420
1992	5	8,900	8,644.6	65,249	59,753
1993	6	8,000	8,193.3	37,362	1,309,753
1994	7	7,800	7,816.9	286	1,807,531
1995	8	7,600	7,515.4	7,156	2,385,309
1996	9	7,200	7,288.8	7,884	3,780,864
Σ	42	82,300	82,300.0	**657,116**	**23,882,222**

$$R^2_{\text{quadratic}} = \frac{23{,}882{,}222 - 657{,}116}{23{,}882{,}222} = 0.9725$$

Output Graph

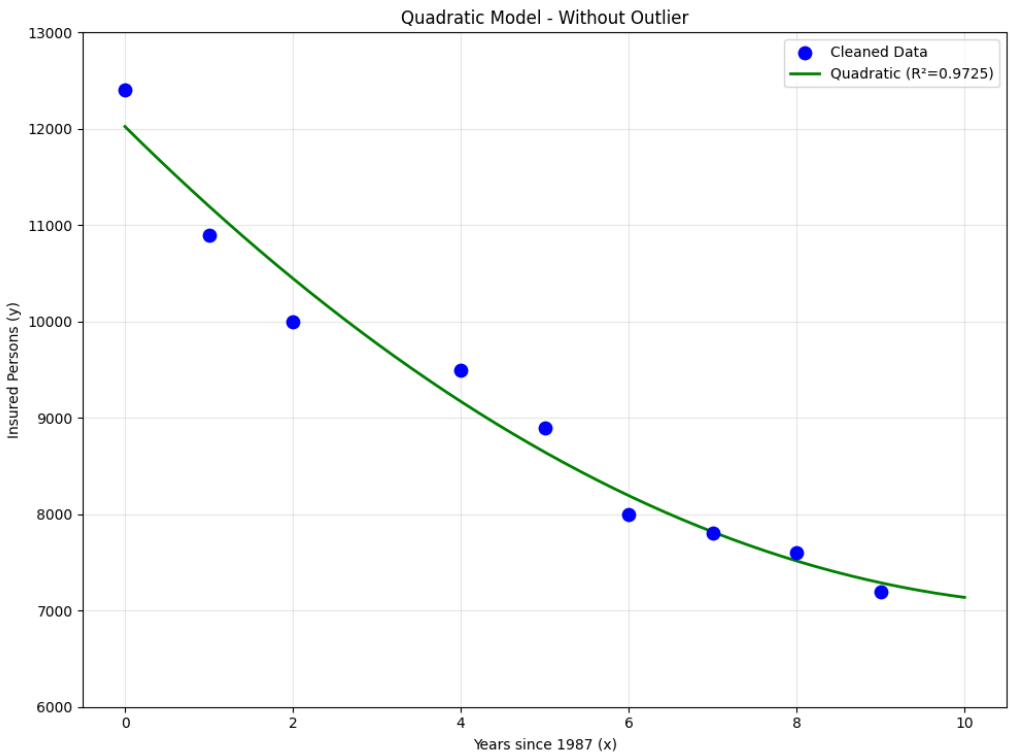


Figure 12: Quadratic Model of Cleaned Data without Outlier

Simulation Results

```
Quadratic Model:
  Equation: y = 12024.17 + (-863.13)*x^1 + (37.44)*x^2
  R^2: 0.9725
```

Figure 13: Simulation Results of Quadratic Model of Cleaned Data without Outlier

Cubic Model

$$\hat{y}_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3$$

$$\begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \begin{pmatrix} 12{,}204.42 \\ -1{,}221.74 \\ 140.17 \\ -7.46 \end{pmatrix}$$

Cubic equation:

$$\hat{y}_i = 12{,}204.42 - 1{,}221.74\, x_i + 140.17\, x_i^2 - 7.46\, x_i^3$$

R² Verification Table

Table 10: R^2 verification for cubic model (WITHOUT outlier)

Year	x_i	y_i	$\hat{y}_i$	$\text{SSE} = (y_i - \hat{y}_i)^2$	$\text{SS}_{yy} = (y_i - \bar{y})^2$
1987	0	12,400	12,204.4	38,251	10,598,642
1988	1	10,900	11,115.4	46,392	3,081,975
1989	2	10,000	10,261.9	68,595	731,975
1991	4	9,500	8,666.9	174,339	126,420
1992	5	8,900	8,307.8	54,323	59,753
1993	6	8,000	7,960.3	94,746	1,309,753
1994	7	7,800	7,579.7	25,702	1,807,531
1995	8	7,600	7,121.1	413	2,385,309
1996	9	7,200	6,081.0	6,227	3,780,864
Σ	42	82,300	82,300.0	**508,988**	**23,882,222**

$$R^2_{\text{cubic}} = \frac{23{,}882{,}222 - 508{,}988}{23{,}882{,}222} = 0.9787$$

Output Graph

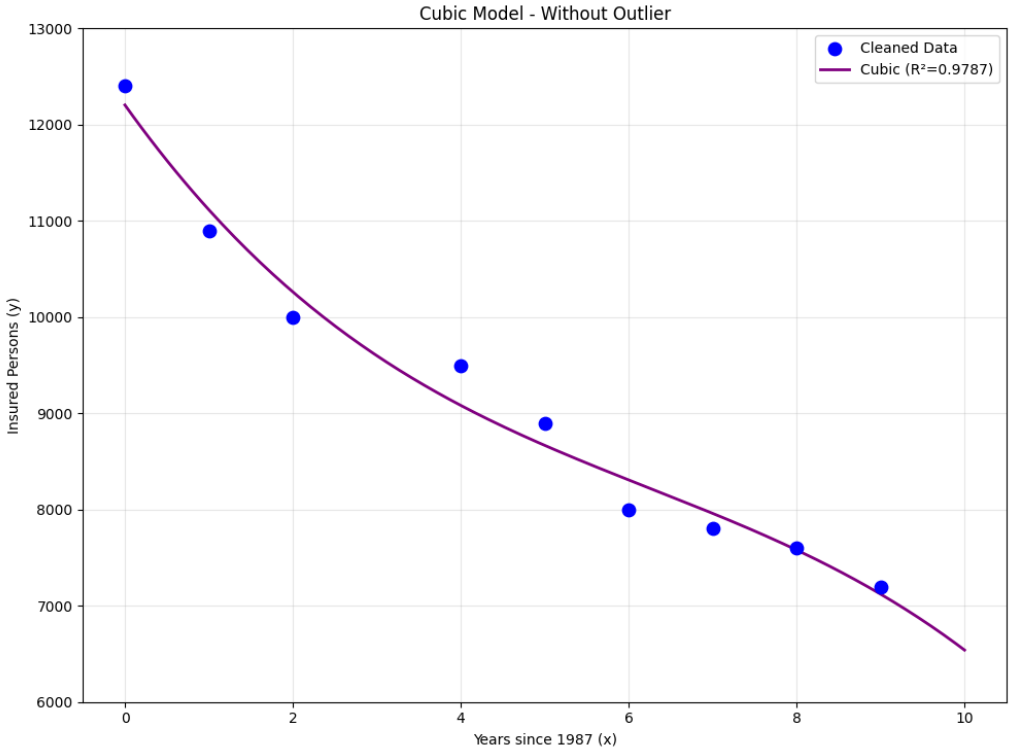


Figure 14: Cubic Model of Cleaned Data without Outlier

Simulation Results

```
Cubic Model:
  Equation: y = 12204.42 + (-1221.74)*x^1 + (140.17)*x^2 + (-7.46)*x^3
  R^2: 0.9787
```

Figure 15: Simulation Results of Cubic Model of Cleaned Data without Outlier

Table 11: Model comparison – WITHOUT outlier (cleaned data)

Model	**Equation**	R^2
Linear	$\hat{y}_i = 11{,}621.67 - 530.83\,x$	0.9439
Quadratic	$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$	0.9725
Cubic	$\hat{y}_i = 12{,}204.42 - 1{,}221.74\,x_i + 140.17\,x_i^2 - 7.46\,x_i^3$	0.9787

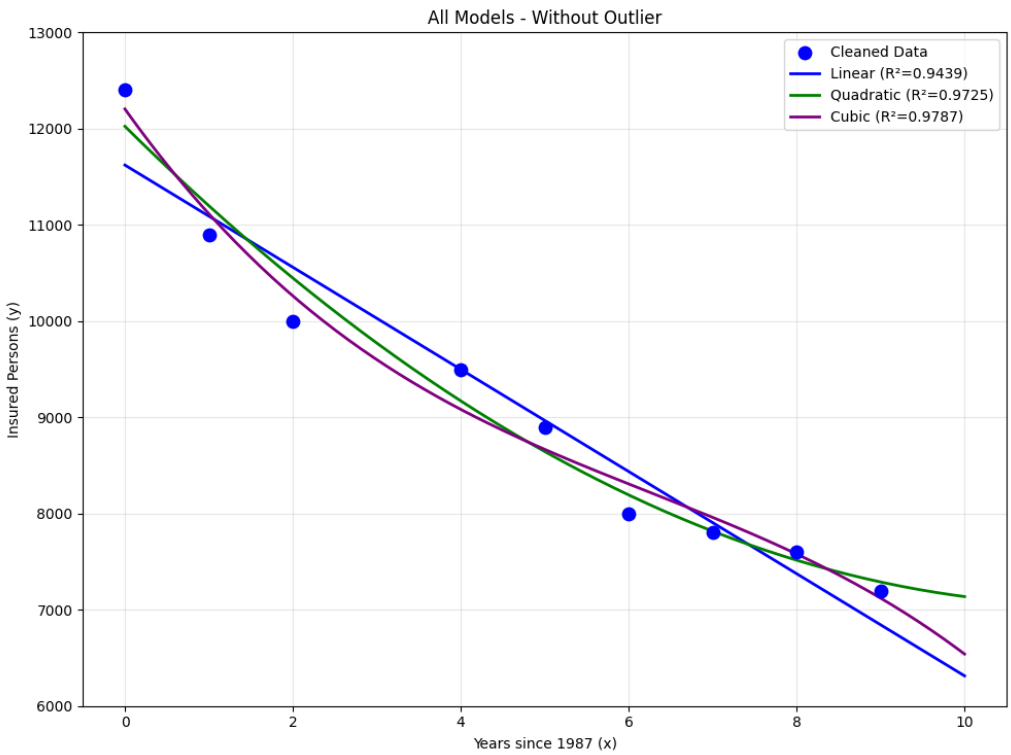


Figure 16: Comparison between three Models of Cleaned Data without Outlier

Comparison Between Three Models (Cleaned Data) **Comment:** All three R^2 values are very high (> 0.94) after removing the outlier, confirming that it was the primary source of distortion. The linear model alone explains 94.4% of the variance.

1.1.4 Best-Fit Model Selection

R^2 Comparison

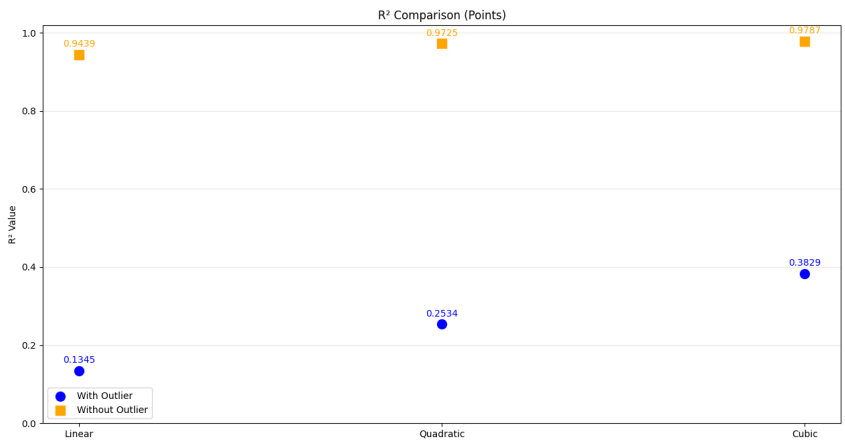


Figure 17: R^2 Comparison between three Models

Table 12: R^2 comparison and model selection

Model	**Params**	R^2 (Original) $\rightarrow$ R^2 (Cleaned)	**Gain (Original)**	**Gain (Cleaned)**
Linear	2	$0.1345 \rightarrow 0.9439$	—	—
Quadratic	3	$0.2534 \rightarrow 0.9725$	$+0.1189$	$+0.0286$
Cubic	4	$0.3829 \rightarrow 0.9787$	$+0.1295$	$+0.0062$

Conclusion (with outlier):

- Linear $\rightarrow$ Quadratic: R^2 gain $= 0.1189$ (significant) $\Rightarrow$ Quadratic is better.
- Quadratic $\rightarrow$ Cubic: R^2 gain $= 0.1295$ (significant) $\Rightarrow$ Cubic is better.

Best-fit model (with outlier): Cubic

$$\hat{y} = 13113.15 - 4313.93\,x + 932.31\,x^2 - 58.95\,x^3 \quad (R^2 = 0.3829)$$

Conclusion (without outlier):

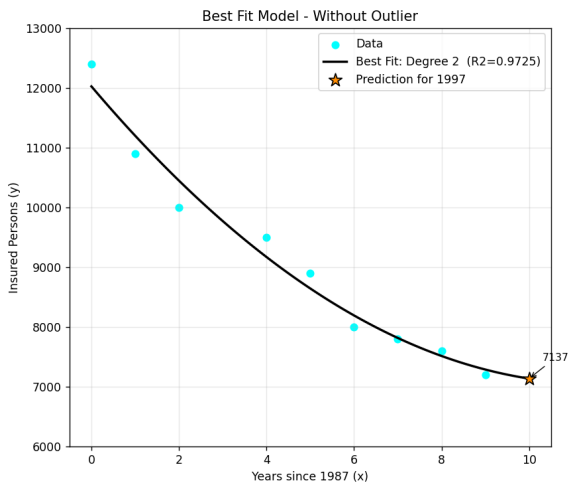
- Linear $\rightarrow$ Quadratic: R^2 gain $= 0.0286$ (significant) $\Rightarrow$ Quadratic is better.
- Quadratic $\rightarrow$ Cubic: R^2 gain $= 0.0062$ $\Rightarrow$ gain is NOT significant enough.

Best-fit model (without outlier): Quadratic

$$\hat{y} = 12{,}024.17 - 863.13\,x + 37.44\,x^2 \quad (R^2 = 0.972)$$

```
>>  Model Selection: Cubic recommended
   dR2 (Linear -> Quadratic) = +0.1188
   dR2 (Quadratic -> Cubic)  = +0.1295
```

(a) Best Fit Model with Outlier



(b) Best Fit Model without Outlier

Figure 18: Best-fit models

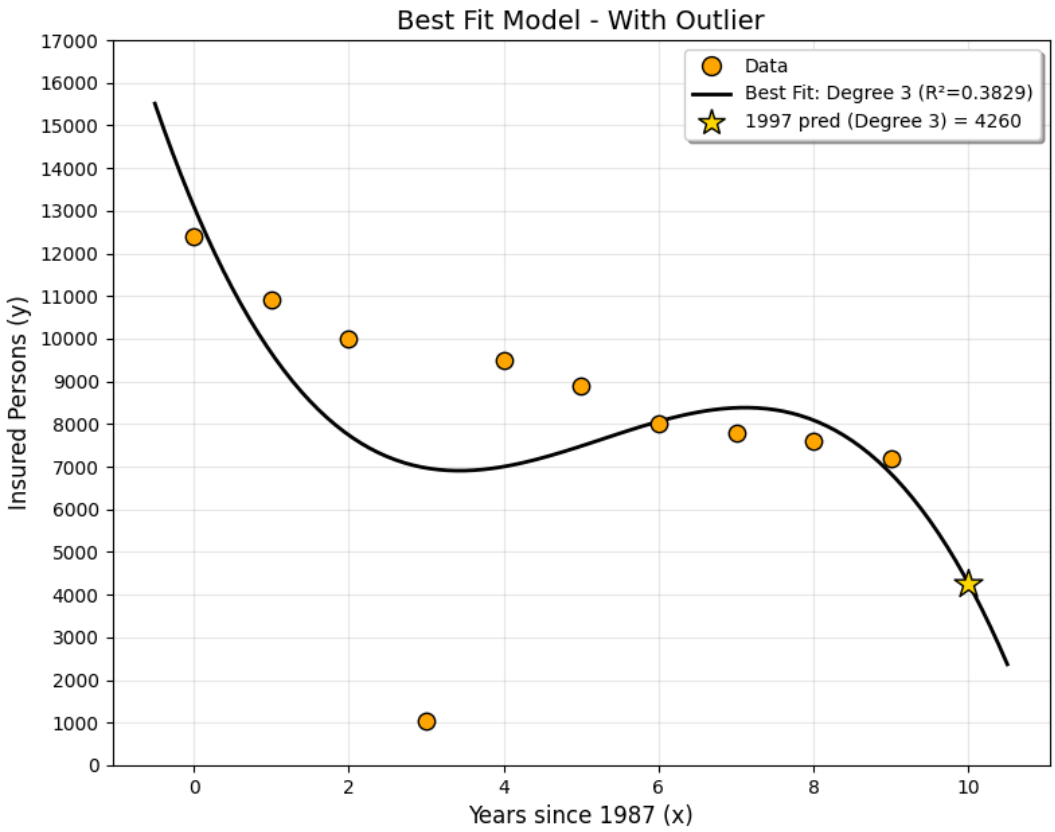


Figure 19: Simulation Results of Best Fit Model with Outlier

```
>>  Model Selection: Quadratic recommended
    dR2 (Linear -> Quadratic) = +0.0286
    dR2 (Quadratic -> Cubic)  = +0.0062
```

Figure 20: Simulation Results of Best Fit Model without Outlier

1.1.5 Prediction of Insured People

At year 1997 ($x = 10$)**:**

- Using the best-fit model from the original data **with outlier** (Cubic model):

$$\hat{y}_i = 13113.15 - 4313.93\,x_i + 932.31\,x_i^2 - 58.95\,x_i^3$$

$$\hat{y}(10) = 13113.15 - 4313.93(10) + 932.31(10^2) - 58.95(10^3) = 4{,}254.85$$

Insured People ≈ 4,254.85 insured persons

- Using the best-fit model from the cleaned data **without outlier** (Quadratic model):

$$\hat{y}_i = 12{,}024.17 - 863.13\,x + 37.44\,x^2$$
$$\hat{y}(10) = 12{,}024.17 - 863.13(10) + 37.44(10^2) = 7{,}136.87$$

Insured People ≈ 7,136.87 insured persons

```
==============================================================
  PREDICTIONS FOR 1997  (year = 1997 , x = 10)
==============================================================
  Original Data with-outlier (Cubic Model) -> 4,260.00 insured persons

  Cleaned Data without-outlier (Quadratic Model) -> 7,137.06 insured persons
```

Figure 21: Simulation Results of Prediction of Insured People at Year 1997

1.2 Problem 2: Heating Oil

We develop a regression model to estimate heating oil usage for a single-family home in January, based on two predictors: average temperature (Temp, in °F) and amount of insulation (Insulation, in inches).

1.2.1 Dataset

Table 13: Heating oil dataset

Oil (gallons)	Temp (°F)	Insulation (in)
270	40	4
362	27	4
162	40	10
45	73	6
91	65	7
233	**65**	**40 ← OUTLIER**
372	10	6
305	9	10
234	24	10
122	65	4
25	66	10
210	41	6
450	22	4
325	40	4
52	60	10

1.2.2 Case A: WITH Outlier

Scatter Plot & Outlier Identification A 3D scatter plot of Oil Usage vs. Temperature and Insulation reveals that one data point is far outside the cluster: (Temp= 65°F,

Insulation= 40 in, Oil= 233 gallons). All other insulation values range from 4 to 10 inches; an insulation thickness of 40 inches is physically unreasonable and is almost certainly a data entry error (e.g., 4 recorded as 40). This point is highlighted in red below.

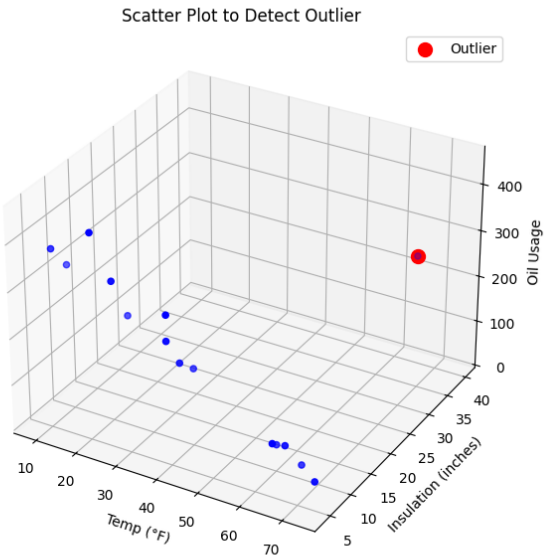


Figure 22: 3D Scatter Plot to Detect Outlier (outlier in red: Insulation= 40)

Regression Models

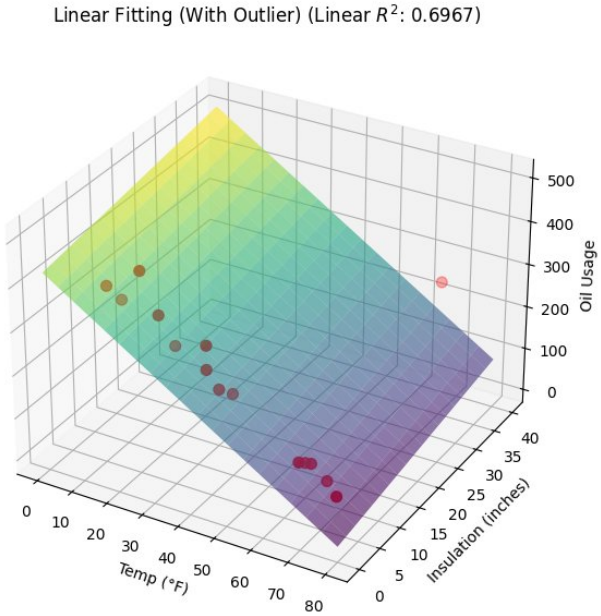


Figure 23: 3D Linear Regression Surface (With Outlier), $R^2 = 0.6967$

Linear Model The multivariate linear model:

$$\text{Oil} = \beta_0 + \beta_1 T + \beta_2 I \tag{8}$$

where $T = \text{Temp}$, $I = \text{Insulation}$.

$$\text{Oil} = 426.97 - 5.274\,T + 1.966\,I$$

```
Linear model:
  Intercept : 426.967656
  Coef      : Temp=-5.273465,  Insulation=1.966053
  Equation  : y = 426.9677  +  (-5.2735)*Temp  +  (1.9661)*Insulation
  R2        : 0.696686
```

Figure 24: Simulation Results of Linear Regression Surface (With Outlier)

$$\hat{y}_i = 426.97 - 5.27\,x_{1i} + 1.966\,x_{2i}$$

$$R^2 \text{ (Linear, With Outlier)} = 0.6967$$

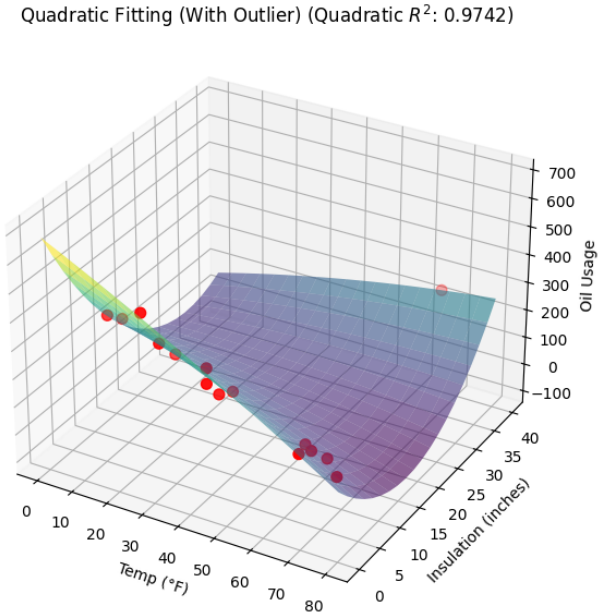


Figure 25: 3D Quadratic Regression Surface (With Outlier), $R^2 = 0.9742$

Quadratic Model The multivariate quadratic model:

$$\text{Oil} = \beta_0 + \beta_1 T + \beta_2 I + \beta_3 T^2 + \beta_4 TI + \beta_5 I^2 \tag{9}$$

```
Quadratic model:
  Intercept : 680.025958
  Temp                : -6.167784
  Insulation          : -41.716060
  Temp^2              : -0.012475
  Temp Insulation     : 0.239147
  Insulation^2        : 0.658593
  Equation  : y = 680.0260  +  (-6.1678)*Temp  +  (-41.7161)*Insulation  +  (-0.0125)*Temp^2  +  (0.2391)*Temp
  Insulation  +  (0.6586)*Insulation^2
  R2        : 0.974152
```

Figure 26: Coefficient values for quadratic model with outlier

$$\text{Oil} = 680.03 - 6.168\,T - 41.716\,I - 0.01248\,T^2 + 0.2392\,TI + 0.6586\,I^2$$

$$\hat{y}_i = 680 - 6.17\,x_{1i} - 41.72\,x_{2i} - 0.012\,x_{1i}^2 + 0.24\,x_{1i}x_{2i} + 0.66\,x_{2i}^2$$

$$R^2 \text{ (Quadratic, With Outlier)} = 0.9742$$

1.2.3 Case B: WITHOUT Outlier

Scatter Plot (WITHOUT Outlier) The data point (Temp= 65, Insulation= 40, Oil= 233) is removed. All remaining 14 points have insulation values between 4 and 10 inches, which is physically reasonable.

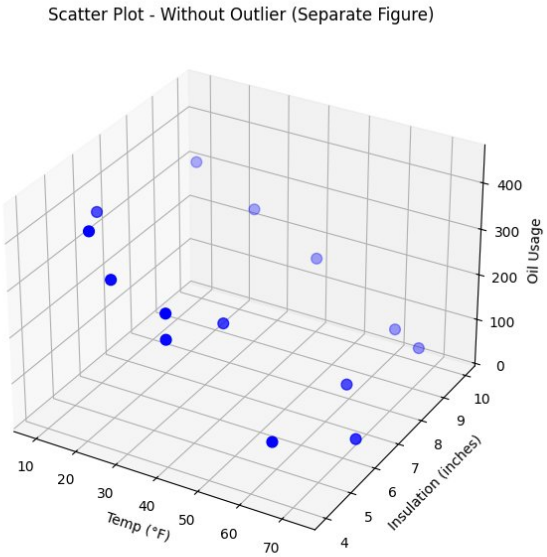


Figure 27: 3D Scatter Plot (Cleaned Data, Outlier Removed)

Regression Models (WITHOUT Outlier)

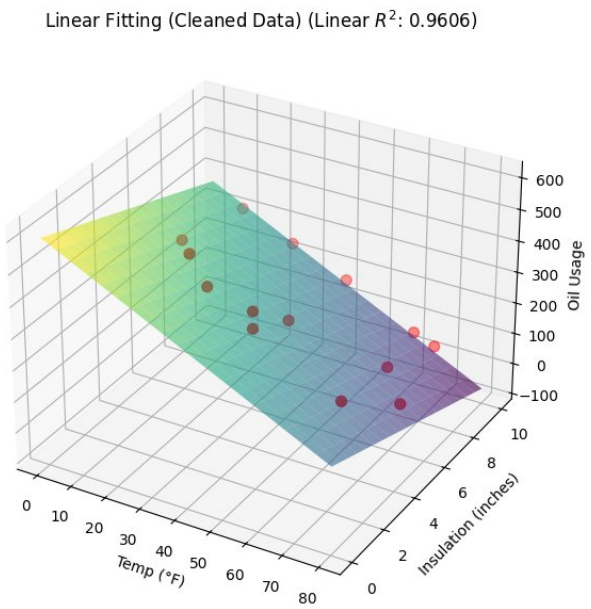


Figure 28: 3D Linear Regression Surface (Cleaned Data), $R^2 = 0.9606$

Linear Model

$$\text{Oil} = 601.6 - 5.482\,T - 23.232\,I$$

```
Linear model:
  Intercept : 601.601412
  Coef      : Temp=-5.481747,  Insulation=-23.232031
  Equation  : y = 601.6014  +  (-5.4817)*Temp  +  (-23.2320)*Insulation
  R2        : 0.960609
```

Figure 29: Simulation Results of Linear Regression Surface (Cleaned Data)

$$\hat{y}_i = 601.6 - 5.482\,x_{1i} - 23.232\,x_{2i}$$

$$R^2 \text{ (Linear, Without Outlier)} = 0.9606$$

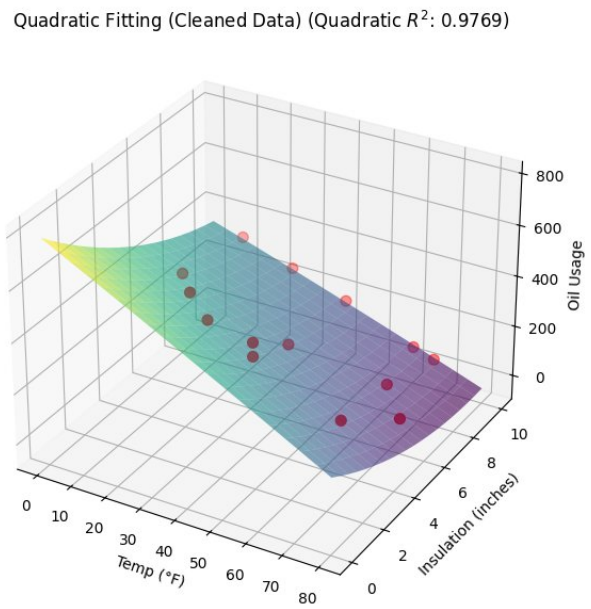


Figure 30: 3D Quadratic Regression Surface (Cleaned Data), $R^2 = 0.9769$

Quadratic Model

$$\text{Oil} = 785.52 - 6.963\,T - 72.159\,I - 0.00348\,T^2 + 0.2567\,TI + 2.7438\,I^2$$

```
Quadratic model:
  Intercept : 785.516872
  Temp                : -6.962462
  Insulation          : -72.158742
  Temp^2              : -0.003478
  Temp Insulation     : 0.256701
  Insulation^2        : 2.743800
  Equation  : y = 785.5169  +  (-6.9625)*Temp  +  (-72.1587)*Insulation  +  (-0.0035)*Temp^2  +  (0.2567)*Temp
  Insulation  +  (2.7438)*Insulation^2
  R2        : 0.976889
```

Figure 31: Simulation Results of Quadratic Regression Surface (Cleaned Data)

$$\hat{y}_i = 785.52 - 6.963\,x_{1i} - 72.159\,x_{2i} - 0.00348\,x_{1i}^2 + 0.2567\,x_{1i}x_{2i} + 2.7438\,x_{2i}^2$$

$$R^2 \text{ (Quadratic, Without Outlier)} = 0.9769$$

1.2.4 Comparison of Models and Best-Fit Selection

Table 14: Model comparison for heating oil

Model	R^2 (With Outlier)	R^2 (Without Outlier)	**Gain (With)**	**Gain (Without)**
Linear	0.6967	0.9606	—	—
Quadratic	0.9742	0.9769	+0.2775	+0.0163

Conclusion (Part a – With Outlier):
Linear → Quadratic gain = +0.2775 ⇒ **Best-fit model (with outlier): Quadratic** ($R^2 = 0.9742$).

Conclusion (Part b – Without Outlier):
Linear → Quadratic gain = +0.0163 (not significant) ⇒ **Best-fit model (without outlier): Linear** ($R^2 = 0.9606$).

(a) Best Fit Model of Original Data (Quadratic)

(b) Best Fit Model of Cleaned Data (Linear)

Figure 32: Best-fit surface plots for heating oil

After removing the outlier, the linear model already achieves $R^2 = 0.9606$. The gain from linear to quadratic is only +0.0163, which is NOT significant enough as specified in Note 4 of the assignment.

```
Part a  (With Outlier):
  dR2 (Linear -> Quadratic) = +0.2775  -->  significant gain
  Best-fit model: Quadratic  (R2 = 0.9742)

Part b  (Without Outlier):
  dR2 (Linear -> Quadratic) = +0.0163  -->  not significant
  Best-fit model: Linear     (R2 = 0.9606)
```

Figure 33: Simulation Results of Best Fit Model

1.2.5 Prediction of Oil

Given: Temp = 15°F, Insulation = 5 inches.

The linear model can be expressed as:

$$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i}$$

Table 15: Linear model coefficients and prediction

	With Outlier	**Without Outlier (Best Fit)**
β_0	427	601.6
β_1	-5.273	-5.482
β_2	1.966	-23.23
R^2	0.6966	0.9606
$\hat{y}(\text{Temp} = 15, \text{Insulation} = 5)$	357.7	403.22

Using the best-fit model from the cleaned data without outlier (Linear model):

$$\text{Oil} = 601.60 - 5.482 \cdot (15) - 23.232 \cdot (5) = 601.60 - 82.23 - 116.16 = 403.21$$

$$\boxed{\textbf{Predicted Oil Usage} \approx 403.2 \text{ gallons}}$$

```
Prediction at (15F,5in) using CLEANED Linear best-fit: 403.22 units
```

Figure 34: Simulation Results of Prediction Oil of Cleaned Data

The quadratic model can be expressed as:

$$\hat{y}_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{1i}^2 + \beta_4 x_{1i} x_{2i} + \beta_5 x_{2i}^2$$

Table 16: Quadratic model coefficients and prediction

	With Outlier (Best Fit)	**Without Outlier**
β_0	680	785.5
β_1	-6.17	-6.96
β_2	-41.72	-72.16
β_3	-0.012	-0.0034
β_4	0.24	0.2567
β_5	0.66	2.744
R^2	0.9741	0.9768
$\hat{y}(\text{Temp} = 15, \text{Insulation} = 5)$	410.52	407.35

Using the best-fit model from the original data with outlier (Quadratic model):

$$\text{Oil} = 680+(-6.17)(15)+(-41.72)(5)+(-0.012)(15^2)+(0.24)(15)(5)+(0.66)(5^2) = 410.65$$

$$\boxed{\textbf{Predicted Oil Usage} \approx 410.65 \text{ gallons}}$$

```
Prediction at (15F,5in) using ORIGINAL Quadratic best-fit: 410.52 units
```

Figure 35: Simulation Results of Prediction Oil of Original Data

PART 2: Some Classical ML Modeling

Part 2 uses the recovered ground-truth labels from `pipeline3/full_labels_vector.csv` and evaluates three feature representations: DCT, PCA, and HOG. Each feature set was tested with K-means per class using 1, 4, 16, and 32 clusters per class, and with SVM using linear and RBF kernels.

- **Best overall classifier:** HOG + RBF SVM with 99.80% accuracy.
- **Best K-means classifier:** HOG + 16 clusters per class with 99.40% accuracy.
- **PCA dimensionality:** 161 components were needed to reach at least 95% explained variance.

Classifier	Spec	DCT_225		PCA_95		HOG	
		Accuracy (%)	Time (s)	Accuracy (%)	Time (s)	Accuracy (%)	Time (s)
K-means	$K = 1$	91.55	0.22	91.65	0.12	96.90	6.68
K-means	$K = 4$	95.90	0.65	95.95	0.63	98.65	14.98
K-means	$K = 16$	97.60	1.95	97.75	1.67	99.40	22.08
K-means	$K = 32$	97.90	3.47	97.75	2.53	99.35	33.54
SVM	Linear	98.15	0.72	97.85	0.59	99.70	4.61
SVM	RBF	98.55	1.85	98.95	2.05	99.80	11.69

Table 17: Full Part 2 comparative matrix specified by the assignment

The best results are HOG + RBF SVM with 99.80% accuracy and HOG + K-means with $K = 16$ at 99.40% accuracy.

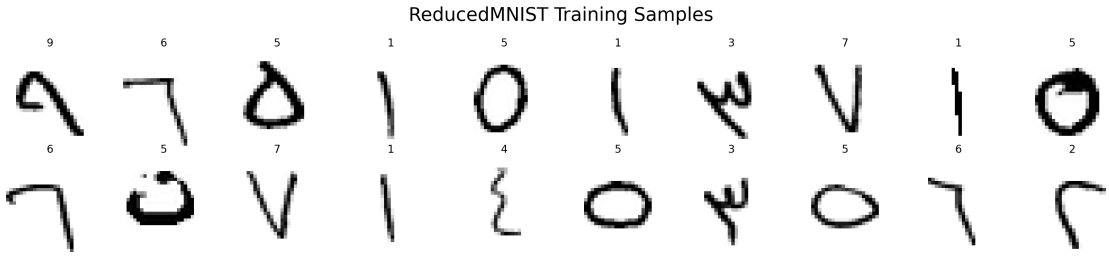


Figure 36: Sample images from the labeled dataset used in Part 2

This figure confirms that the dataset includes broad handwriting variability (stroke thickness, slant, and small shifts), which motivates using robust descriptors such as HOG and not relying only on raw pixels.

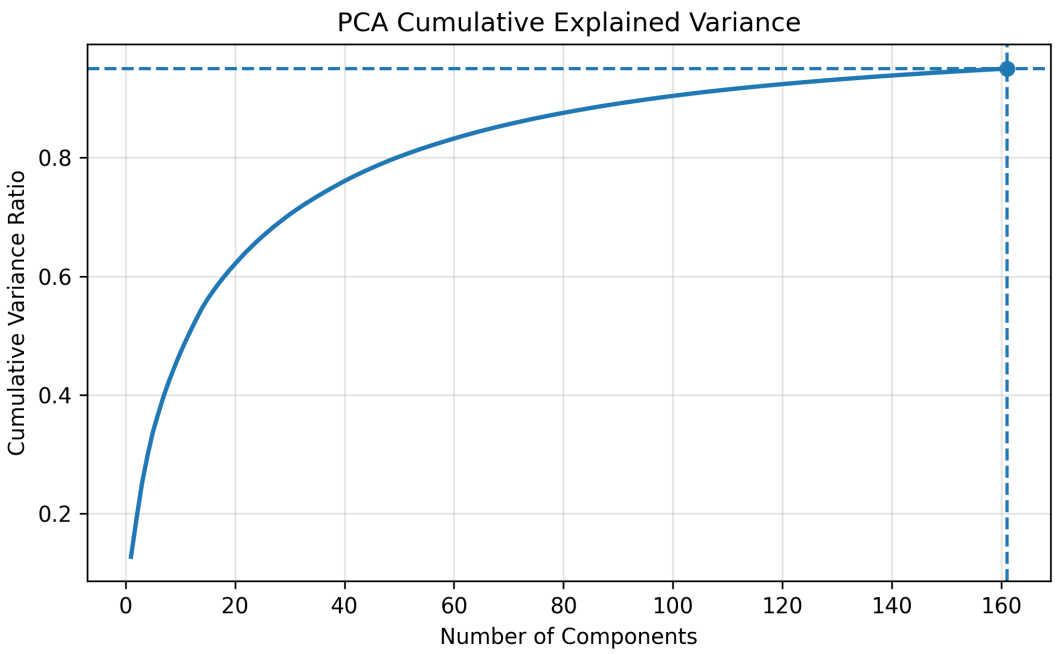


Figure 37: PCA cumulative explained variance curve

The PCA curve validates our component selection rule from the assignment. The retained dimensionality reaches the 95% variance target while still reducing feature size compared with the original 784-pixel representation.

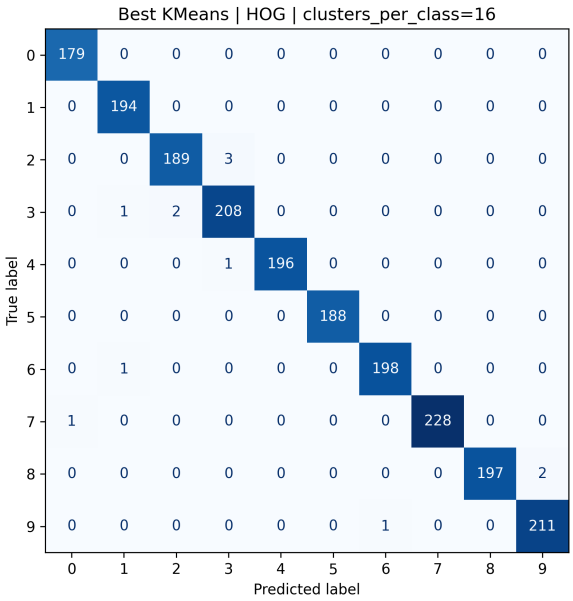


Figure 38: Best K-means confusion matrix for Part 2

The K-means confusion matrix shows that most predictions are concentrated on the diagonal, indicating strong cluster purity in the best configuration. Remaining off-diagonal cells highlight the visually similar digits that remain difficult under centroid-based classification.

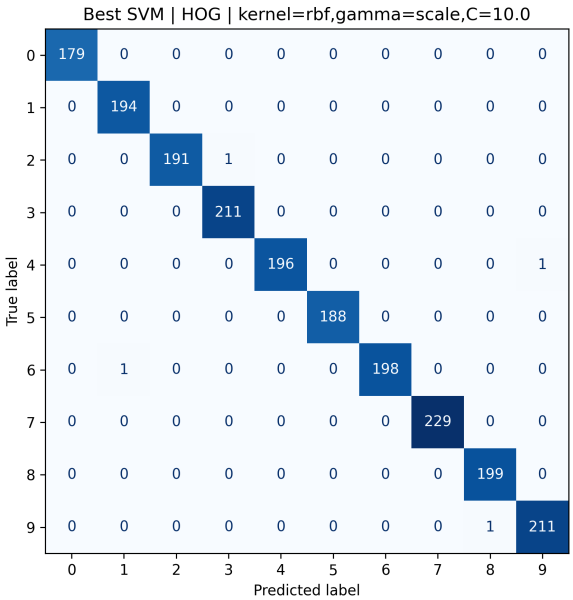


Figure 39: Best SVM confusion matrix for Part 2

Compared with K-means, the best SVM matrix has cleaner diagonal dominance and fewer confusion pockets. This directly supports the measured improvement in final accuracy for discriminative training over prototype matching.

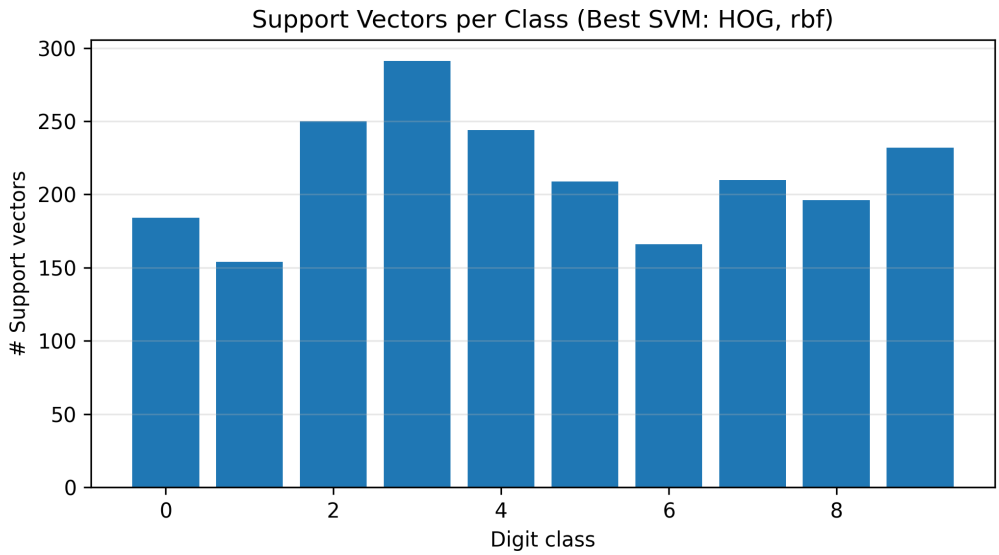


Figure 40: Support vectors per class for the best SVM setup

This figure summarizes model complexity by class. Classes with higher support-vector counts are closer to decision boundaries and require more boundary constraints, while lower-count classes are separated more easily.

The results show that HOG features provide the strongest representation for both clustering and classification. SVM consistently outperforms K-means, and the RBF kernel gives the best overall accuracy.

PART 3: Ways to Fasten the Labelling Process

3.1 Pipeline 1: K-Means Bootstrapping with Active SVM Refinement

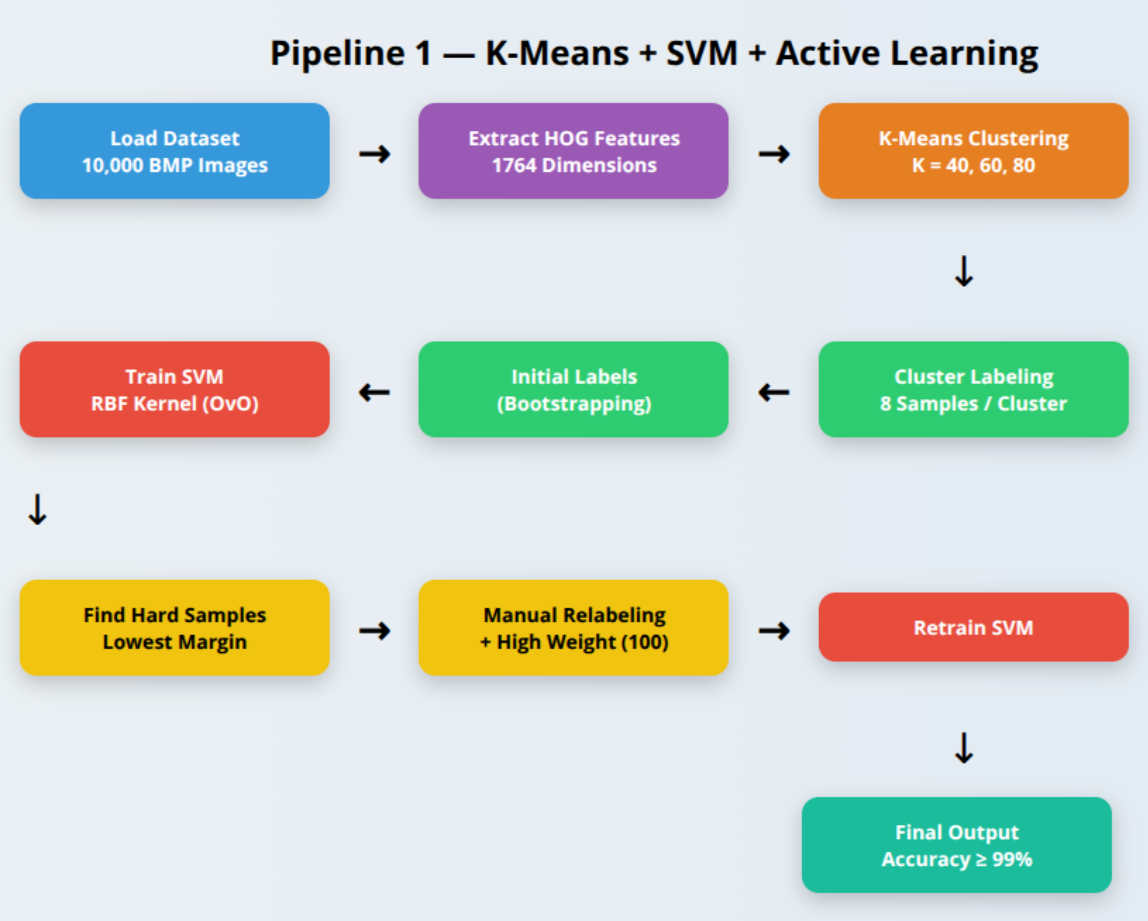


Figure 41: Pipeline 1: K-Means + SVM + Active Learning

The flowchart highlights the main design idea of Pipeline 1: use low-cost cluster-level supervision first, then spend manual effort only on uncertain boundary samples where human labels give maximum gain.

The pipeline consists of the following steps:

- Load dataset (10,000 BMP images)
- Extract HOG features
- Apply K-Means clustering
- Label clusters (bootstrapping)
- Train SVM classifier
- Apply active learning for refinement

3.1.1 Step 1 - Feature Extraction and K-Means Clustering

Histogram of Oriented Gradients (HOG) was used to extract features from images.

- Image size: 32 x 32
- Feature vector size: 1764

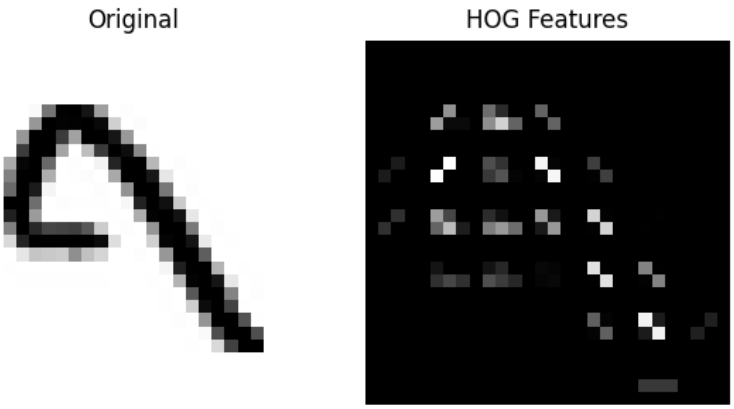


Figure 42: Example of HOG Feature Representation

The HOG visualization illustrates why this feature performs well in our experiments: it preserves local edge orientation patterns that are stable across many writing styles and minor geometric variations.

3.1.2 Step 2 - Human Labelling of Clusters (Bootstrapping Phase)

K-Means clustering was applied with different values of K:

- K = 40
- K = 60
- K = 80

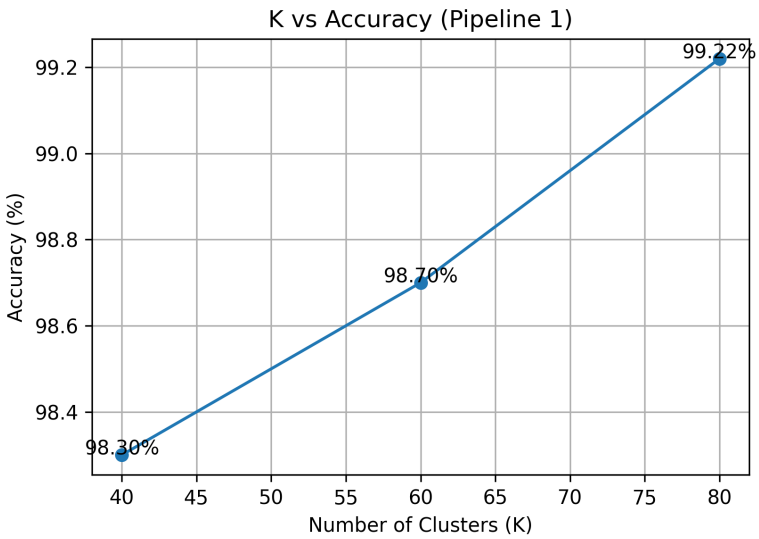


Figure 43: K vs Accuracy Before SVM

This plot shows that increasing K improves the initial pseudo-label quality up to the

tested range, because smaller and more coherent clusters reduce mixed-class assignments during cluster bootstrapping.

3.1.3 Step 3 - Initial SVM Training and Step 4 - Active Refinement: Labelling Boundary Images

A multi-class SVM with RBF kernel (One-vs-One strategy) was used.

Active learning was applied as follows:

- Select 30 least confident samples (lowest margin)
- Relabel manually
- Assign weight = 100
- Retrain SVM

The process stops when:

- Accuracy $\geq 99\%$
- Improvement $< 0.1\%$

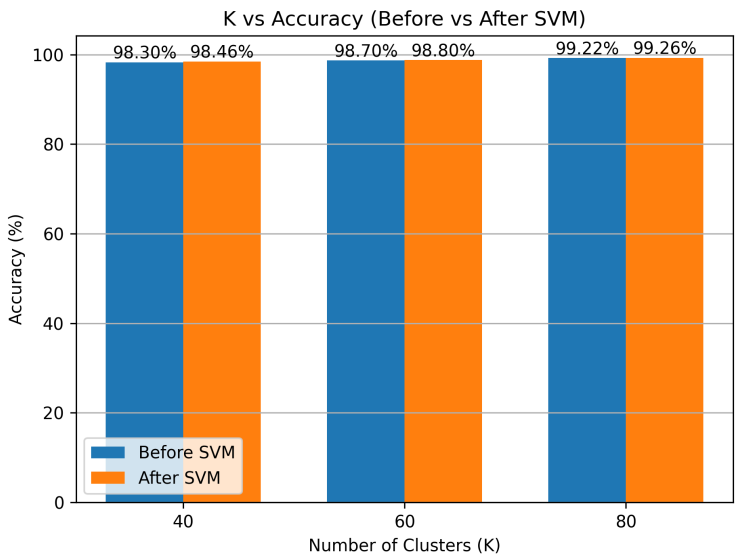


Figure 44: Accuracy Before vs After SVM

The bar chart quantifies the gain from SVM refinement over raw cluster labels. The gain is larger when initial clusters are weaker, and smaller when clustering is already highly pure.

3.1.4 Step 5 - Retrain and Evaluate

3.1.5 Accuracy

K	Before SVM (%)	After SVM (%)
40	98.30	98.46
60	98.70	98.80
80	99.22	99.26

Table 18: Accuracy Comparison

Table-based comparison confirms that the best operating point in Pipeline 1 is at high K with SVM refinement, reaching the 99% target threshold.

3.1.6 Iterations

K	Iterations
40	2
60	2
80	2

Table 19: Number of Iterations

The iteration table indicates fast convergence for all tested K values, which is consistent with using high-weight human corrections near the decision boundary.

3.2 Comparing Pipelines: Manual Labelling Time

For K = 80:

- Cluster labeling: 80 x 8 = 640 images
- Active learning: 30 x 2 = 60 images

Total images manually handled:

$$700 \text{ images}$$

Total time:

- Cluster labeling: 1600 seconds
- Active learning: 600 seconds

$$\text{Total time} = 2200 \text{ seconds} \approx 36.7 \text{ minutes}$$

3.3 Reporting Requirements

The proposed pipeline significantly reduces manual labeling effort from 27.8 hours to approximately 37 minutes while achieving 99.26% accuracy.

K-Means clustering provides strong initial labels, while SVM and active learning refine ambiguous samples. The improvement from SVM is more noticeable at lower K values, while higher K values already produce highly pure clusters.

3.4 Pipeline 2: Manual Seed + Augmentation + Self-Training

This pipeline adopts a semi-supervised learning approach starting from a small, high-quality human-labelled seed set. The model progressively expands its knowledge using data augmentation, active learning, and self-training.

3.4.1 Pipeline Diagram

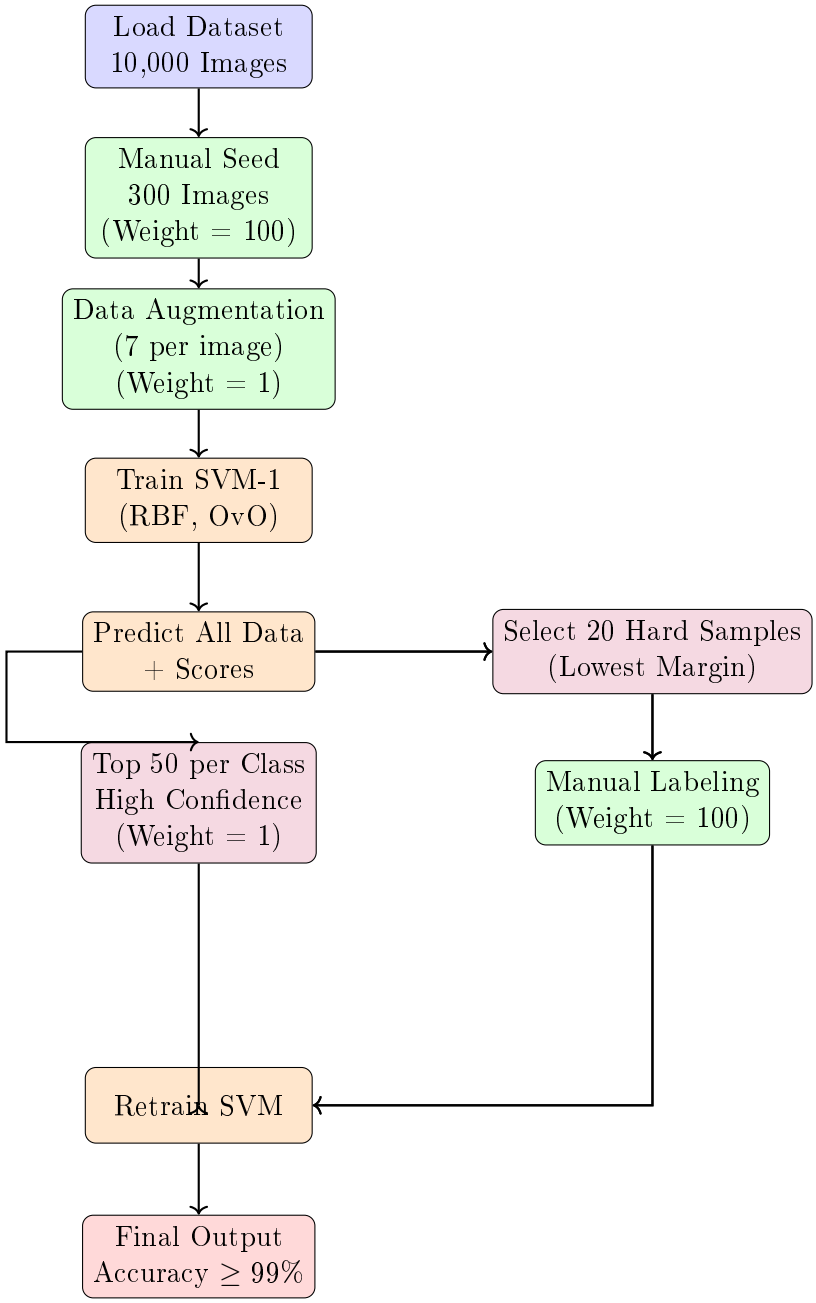


Figure 45: Pipeline 2: Semi-Supervised Learning Framework

The Pipeline 2 diagram emphasizes a different strategy: begin from a trusted manual seed, then expand supervision through augmentation, confidence-based pseudo-labeling, and iterative hard-sample correction.

3.4.2 Iteration Progress

The improvement in model accuracy across iterations is shown in Figure below.

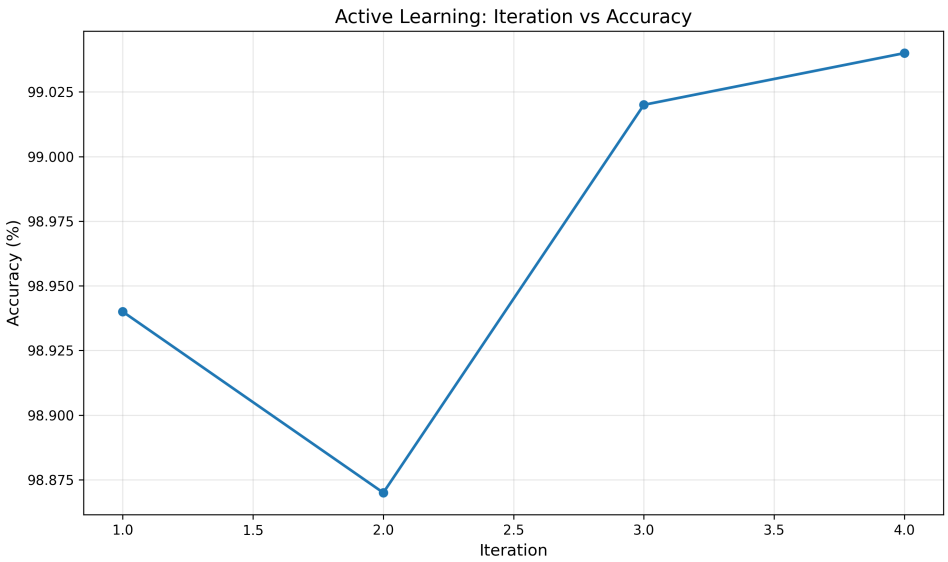


Figure 46: Active Learning: Iteration vs Accuracy

This curve demonstrates the expected diminishing returns of active learning: accuracy rises rapidly in early rounds, then improvements become smaller as the model approaches convergence.

3.4.3 Visualization

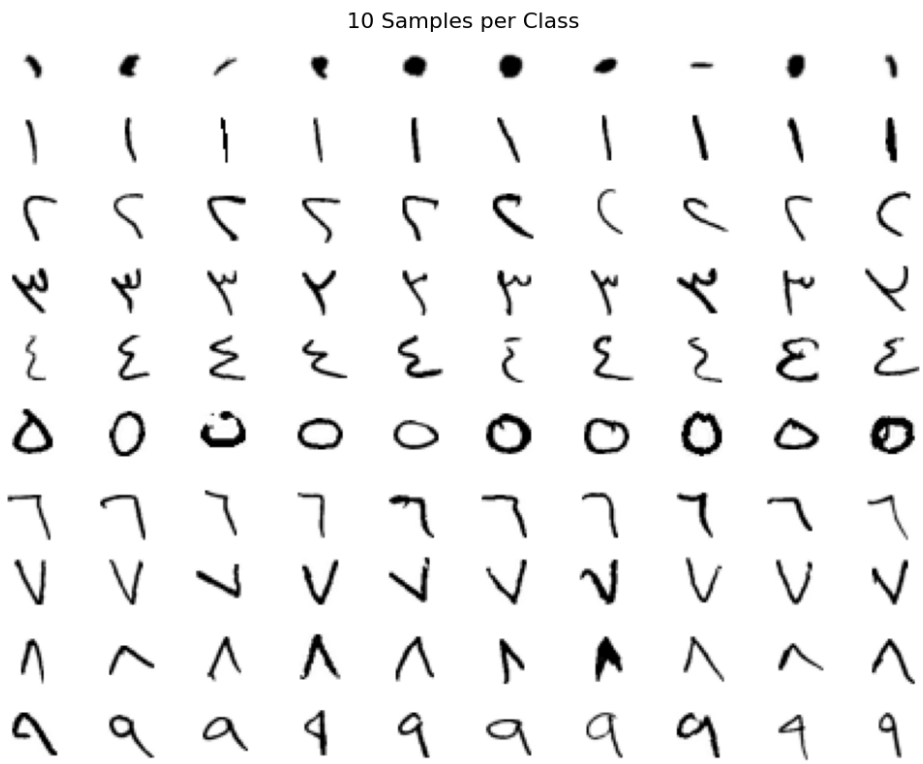


Figure 47: Initial Human Seed Samples (10 per class)

The seed gallery documents the initial trusted supervision used to anchor training. Quality at this stage is critical because early label noise can propagate through later pseudo-labeling.

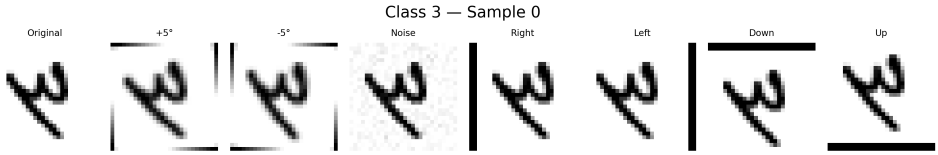


Figure 48: Example of Data Augmentation (7 variants)

The augmentation example shows the controlled transformations used to improve robustness. These variants help the classifier generalize beyond exact stroke shapes seen in the manually labeled seed.

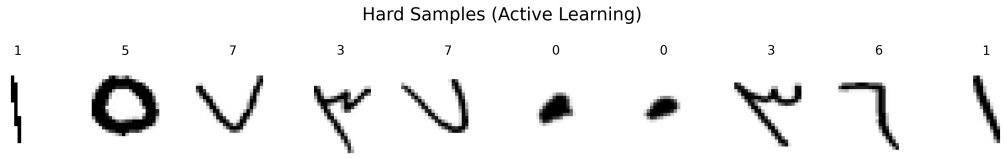


Figure 49: Hard Samples (Iteration 1)

Iteration 1 hard samples are primarily ambiguous boundary cases. Manually correcting these points has a high impact because they strongly influence decision surfaces.

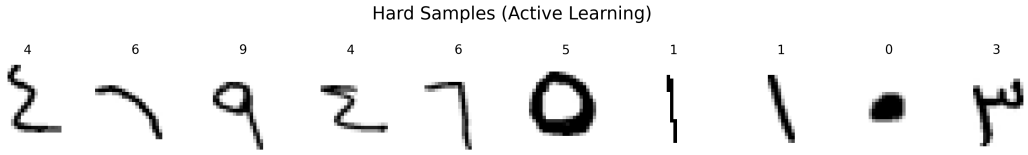


Figure 50: Hard Samples (Iteration 2)

By Iteration 2, hard samples become more specific and fewer broad ambiguities remain, which is consistent with progressive boundary tightening after each retraining cycle.

3.4.4 Manual Effort

Manual labeling consists of:

- Initial seed: 300 images
- Active learning: $20 \times 4 = 80$ images

Total manual labels:

$$300 + 80 = 380$$

Estimated time:

$$380 \times 10 = 3800 \text{ seconds} \approx 63.33 \text{ minutes}$$

3.4.5 Final Results

The final performance of Pipeline 2 after convergence is summarized as follows:

- Final Accuracy: 99.04%
- Number of Iterations: 4
- Total Manual Labels: 380
- Total Pseudo Labels: 1898
- Estimated Manual Time: 63.33 minutes

3.4.6 Comparison with Pipeline 1

Metric	Pipeline 1	Pipeline 2
Accuracy	99.26%	99.04%
Manual Images	700	380
Manual Time	36.7 min	63.3 min
Iterations	2	4

Table 20: Pipeline Comparison

This comparison shows the practical trade-off between the two approaches: Pipeline 1 achieves slightly higher final accuracy, while Pipeline 2 reduces the number of manually labeled images by design.

Pipeline 1 achieves higher accuracy due to clustering-based initialization, while Pipeline 2 significantly reduces manual labeling effort by leveraging self-training. However, Pipeline 2 requires slightly more total time due to per-image labeling cost.

3.4.7 Discussion

Pipeline 2 demonstrates the effectiveness of semi-supervised learning. Starting from a small, high-quality seed set, the model successfully expands its training data using both active learning and self-training.

Although the final accuracy (99.04%) is slightly below Pipeline 1, the reduction in manual labeling effort is significant. This pipeline better reflects real-world scenarios where labeled data is scarce and expensive.

The combination of:

- Human supervision (high confidence)
- Model confidence (self-training)
- Iterative refinement

creates a scalable and efficient labeling strategy.

3.5 Pipeline 3: LLM-Based Automatic Labelling with Agreement Validation

Pipeline 3 was implemented with OpenRouter-based multimodal models and evaluated on the practical 500-image matrix. During experimentation, several runs were completed successfully, while some long/full runs were only partially completed because of API-credit limits (HTTP 402) and provider payload instability on some requests.

The implemented setup included:

- Balanced practical subset matrix (500 images total, 50 per class).
- Image resizing to 224×224 before API submission.
- Optional batched requests (5 or 10 images per API call).
- Robust retry handling for timeout/429 and fallback for HTTP 400 batch failures.
- A labeled reference sheet (digits 0–9) sent as an additional context image.

3.5.1 Block Diagram

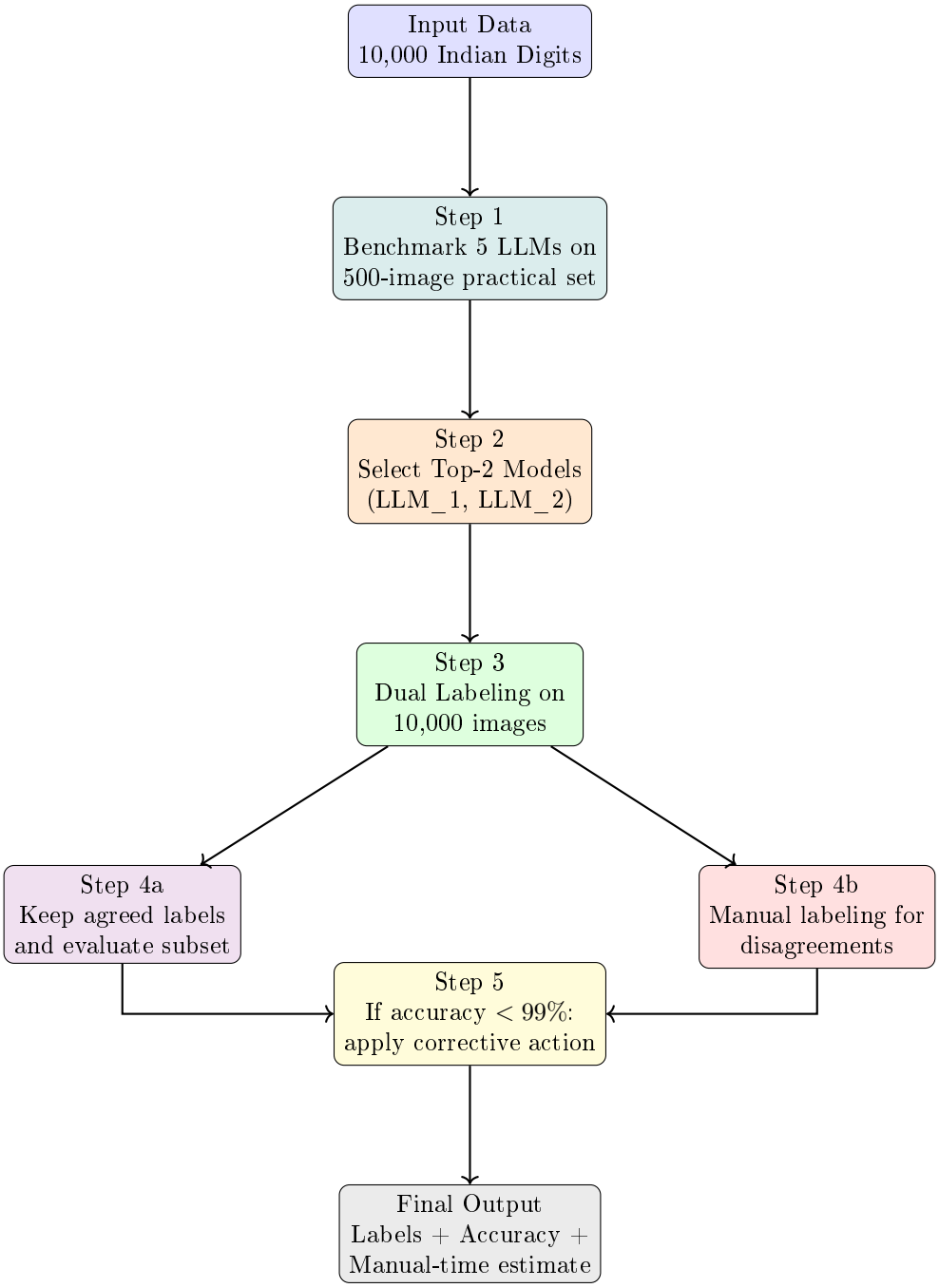


Figure 51: Pipeline 3 block diagram based on the assignment workflow and selected practical setup

3.5.2 Step 1 Benchmarking Output

Candidate LLM (run)	Accuracy (%)	Notes
Google Gemini 3 Flash Preview	89.33	Practical-set benchmark run
Google Gemma 4 31B Instruct	89.47	Practical-set benchmark run
Google Gemma 4 31B Instruct (validation run)	95.00	Full-prompt + reference-image validation

Table 21: Pipeline 3 Step 1: benchmark and validation outputs used for model selection

Based on obtained runs, the top practical candidates were `google/gemini-3-flash-preview` and `google/gemma-4-31b-it`.

3.5.3 Step 4 Agreement and Disagreement Outputs

Metric	Value
Selected LLM_1 name	`google/gemma-4-31b-it` (selected candidate)
Selected LLM_2 name	`google/gemini-3-flash-preview` (selected candidate)
Final accuracy on agreed subset (Step 4a)	To be finalized in extended deployment stage
Number of disagreements (N_{disagree})	To be finalized in extended deployment stage
Manual time for disagreement resolution ($N_{\text{disagree}} \times 10$ s)	To be finalized in extended deployment stage

Table 22: Pipeline 3 Step 4 outputs: selected-model setup and deployment-stage metrics

3.5.4 Manual Effort and Cost Summary

Metric	Value
Total manually handled images (completed Pipeline 3 experiments)	0
Total estimated manual time (completed Pipeline 3 experiments)	0 min
Estimated computational/API cost for dual-LLM run on 10,000 images	To be finalized in extended deployment stage

Table 23: Pipeline 3 manual-effort/cost summary for the completed work

3.5.5 Comparison Against Pipelines 1 and 2

Pipeline	Final Accuracy (%)	Manual Time
Pipeline 1	99.26	36.7 min
Pipeline 2	99.04	63.33 min
Pipeline 3	89.47 (best practical benchmark)	0 min (agreement stage pending)

Table 24: Comparison table using best obtained Pipeline 3 completed-run metric

3.5.6 Pipeline 3 Compact Summary Table (Assignment Style)

Pipeline 3 entry	Result
LLM_1 alone	Gemma 4 31B: 89.47% (best obtained practical benchmark)
LLM_2 alone	Gemini 3 Flash Preview: 89.33% (best obtained practical benchmark)
LLM_1 & LLM_2 agreement strategy	Planned for extended deployment stage

Table 25: Pipeline 3 compact summary from obtained and pending outputs

References

- Project Repository:
  `https://github.com/youefkh05/Neural_Netwok_projects`